Matt Lasselle

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Education:

Iowa State University, College of Engineering Expected Graduation: May 2026

Bachelor of Science, Mechanical Engineering |Dean's List Spring 2023, 2024, 2025 GPA: 3.42

Professional Experience:

Stratasys – Test Engineer Intern *May 2025 – August 2025*

- Developed and programmed a compression test method on a MTS Criterion Model 43 using TestSuite TW Elite, enabling improved data logging and more accurate measurement of a wide range of material properties
- Assessed material data of 3D printed polymers to establish new metrics for defining compressive yield strength to improve data provided to customers
- Reduced required compression testing specimens by analyzing the effect of slenderness ratio on compressive yield strength, while maintaining ASTM D695-10 compliance
- Automated data formatting to increase post-test analysis efficiency and implemented safety measures in test methods to improve lab safety and protect equipment and operators

Technical Projects:

IREC 2026 Competition Rocket *June 2025 – Present*

- Led the design and simulation of the competition flight vehicle consisting of over 500 independent components
- Performed load frame testing on composite and additively manufactured components to evaluate tensile, compressive, bending, high-temperature, and creep failure behavior
- Fabricated subsystem components from aluminum, stainless steel, composites, and plastics using machining, additive manufacturing, and molding processes
- Developed two test vehicles to serve as primary platforms for validating and iterating competition rocket subsystems

Dual Extrusion 3D Printer *August 2025 – Present*

- Designed and manufactured a modular dual-extrusion 3D printer that uses a servo-actuated secondary nozzle
- Performed fluid and thermal analysis on the toolhead incorporating copper, stainless steel, and ABS components
- Validated interlayer bonding strength between materials printed from each nozzle through tensile testing

Solid Rocket Engine and Test Stand *November 2025 – January 2026*

- Modeled a solid rocket motor and test stand to measure thrust, chamber pressure, and casing temperature during static fire testing
- Validated combustion chamber strength and seals through hydrostatic testing up to 1.5 times the expected pressure
- Developed an additively manufactured nozzle incorporating a graphite throat to mitigate erosion

Mechanical Load Frame *May 2025 – November 2025*

- Designed and built a mechanical load frame capable of recording load, displacement, and time data, with a peak capacity of 1,200 lbf and 6.5" of travel
- Wired and programmed the electrical system, including stepper motors, motor drivers, OLED displays, load cells, control buttons, and Arduino microcontrollers

Piston Ejection System *June 2024 – June 2025*

- Designed a 2" diameter piston with focus on failure modes produced from over-pressurization of the piston cylinder
- Validated piston system through hydrostatic testing to a proof of 2 times the expected operating pressure
- Fabricated piston components from aluminum utilizing the mills and lathes

Leadership:

Vice President of Engineering of Cyclone Rocketry Club *June 2025 – Present*

- Oversaw all engineering activities for the club, reviewed and approved critical design decisions, and led the development of subsystem technical documentation and reports
- Served as the flyer of record, ensured full competition compliance, and led system-level testing and validation

Mechanical Team Lead of Cyclone Rocketry Club *June 2024 – June 2025*

- Led a team of 5 students in designing and testing a piston ejection system to separate a 70 lb rocket and deploy the main parachute.

Technical Skills and Certifications:

Pressure Vessel Design	Compressive Material Testing	Data Analysis	SolidWorks	ANSYS Mechanical	Aluminum and Steel Machining
FDM 3D Printing	Fusion 360 CAM	DC Electronics	Klipper	Excel	Tripoli L3 Certified